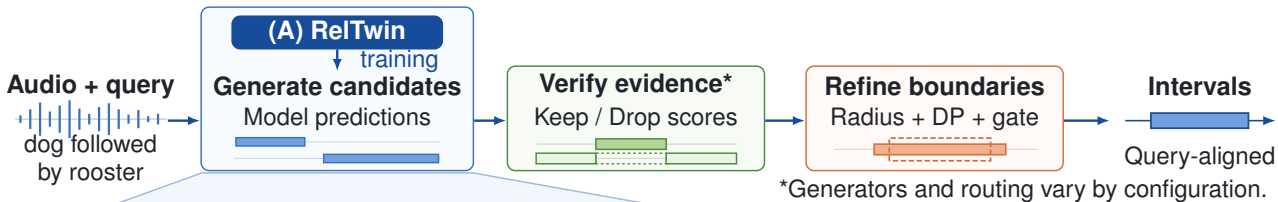
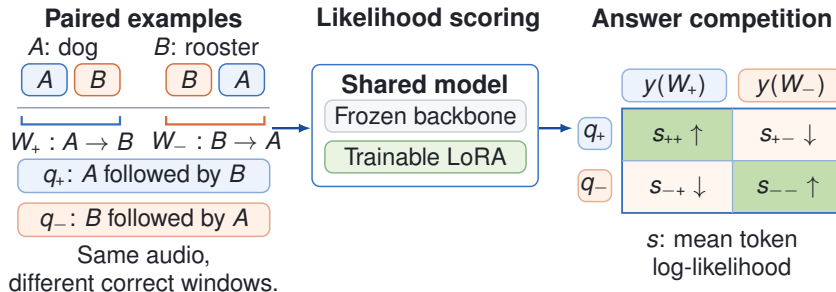


NOVA | System overview



RelTwin module expanded below

(A) RelTwin | Module detail



Training objective

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}$$

$\mathcal{L}_{\text{cand}}$: row-wise cross-entropy

Matched $\uparrow$ Swapped $\downarrow$

Localization example

Same audio; inverse queries

$q_+ : A \rightarrow B$ IoU

Target

SFT 0.00

RelTwin 0.69

$q_- : B \rightarrow A$ IoU

Target

SFT 0.98

RelTwin 0.98

Time (s) 0 10 20 27